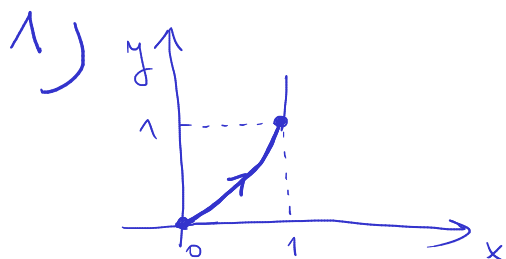


Bonus 5 - RESENU!



$$x=t \Rightarrow dx=dt$$

$$y=t^2 \Rightarrow dy=2t dt \quad , t \in [0,1]$$

$$\int_0^1 (t^2 + 1 + (t^2 - t^2) \cdot 2t) dt = \int_0^1 (t^2 + 1) dt = \left[\frac{t^3}{3} + t \right]_0^1 = \frac{1}{3} + 1 = \underline{\underline{\frac{4}{3}}}$$

$$\begin{array}{l} 2) \quad P = 3x^2 + y \quad \Rightarrow \quad \frac{\partial P}{\partial y} = 1 \\ \quad Q = x - 2y \quad \Rightarrow \quad \frac{\partial Q}{\partial x} = 1 \end{array} \quad \left. \vphantom{\begin{array}{l} P = 3x^2 + y \\ Q = x - 2y \end{array}} \right\} \Rightarrow \text{NEZÁVISLÍ}$$

$$\bar{\Phi} = \int (3x^2 + y) dx = x^3 + xy + c_1(y)$$

$$\bar{\Phi} = \int (x - 2y) dy = xy - y^2 + c_2(x)$$

$$\Rightarrow \underline{\underline{\bar{\Phi} = x^3 + xy - y^2}}$$

$$\bar{\Phi}(1,2) - \bar{\Phi}(0,0) = 1 + 2 - 4 = \underline{\underline{-1}}$$